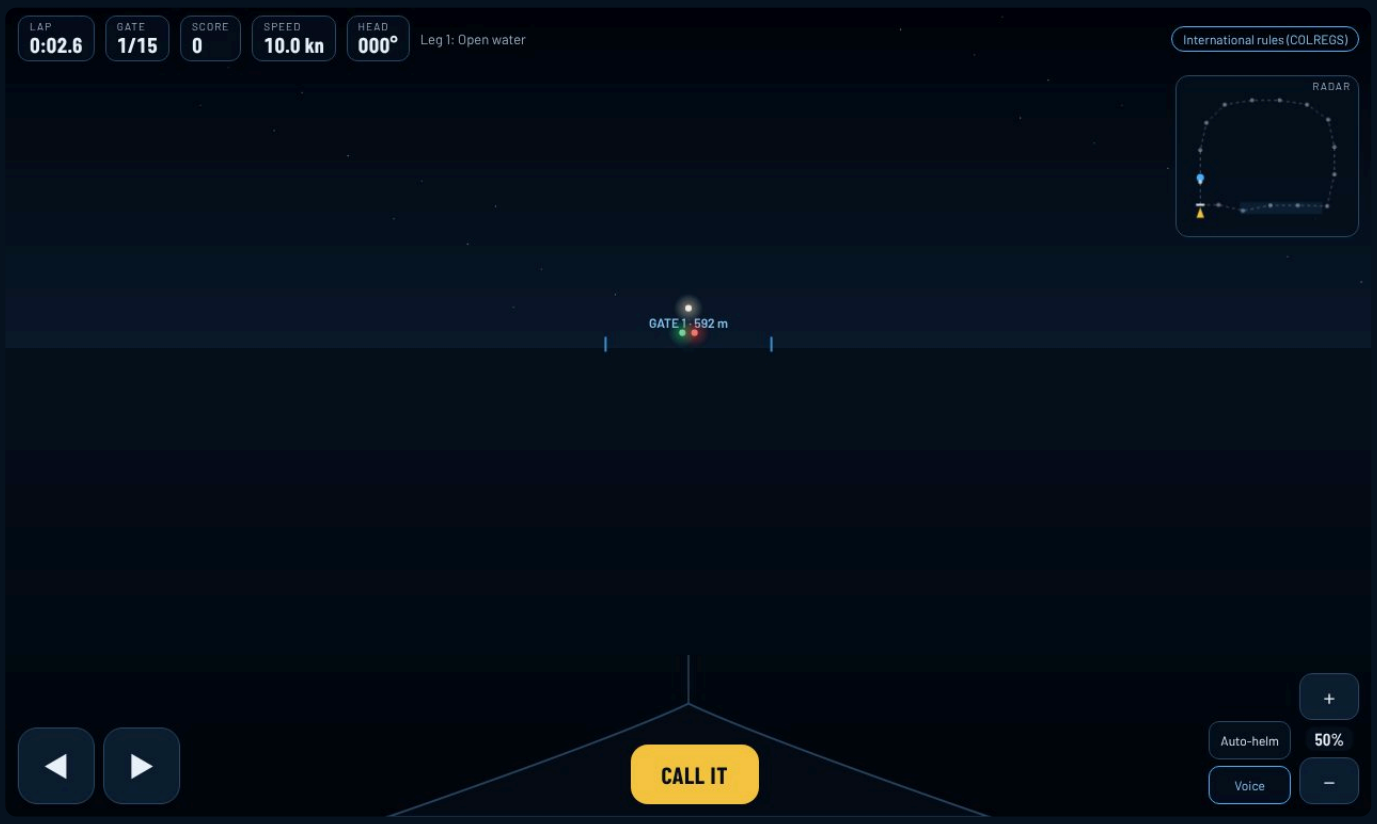
CAPTAIN'S ACADEMY · ARCADE GAME · ENGINEERING HANDOFF

Rules of the Road game: design, prototypes and build

Everything from the 21 to 22 October 2026 design and build session: the game map, every mockup, the playable night prototype, the two levels built into the repo, annotated wireframes, data tables, verification, and what is still open.



Night Harbor Run: a power-driven vessel head-on, only the lights whose arcs face you are drawn

Repository: github.com/Etdev2/Captains-academy-
Prepared by Claude (Cleat Transom) for Et · Session session_01DZNZCfsHuaR57FZWbDhTeB
Status: two branches built and pushed, no pull requests opened yet. Rules content awaits the uscg-regs check.

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1. Status at a glance

Piece	Where it lives	State	Ticket
Docking levels 1 and 2 at /play	`main`	Merged before this session	#25, #26
Level 3: harbor course with traffic	`head-dev/harbor-traffic-course`	Built, verify.sh green, proof committed, no PR	#46
Chapter 1, level 1: Definitions at /rules	`head-dev/ch1-definitions`	Built, verify.sh green, proof committed, no PR	#47
Night Harbor Run (single HTML file)	Private Claude artifact and the zip sent with this PDF	Prototype, approved by Et, not in repo	#45
Chapter 1 mockups, decks, course design	Private Claude design canvas; images in this PDF	Mockups	-
Gallery page and Grok handoff	Private Claude artifact; Markdown file	Documents	-

> Merge note: both branches edit `.claude/skills/verify-captains-academy/drive.mjs`, its `SKILL.md` and `features/README.md`. Whichever merges second needs those files merged by hand and a fresh proof bundle. Keep `LEVER_HALF = 135` from #46.

2. How the conversation went

The requests in order, and what each produced.

#	Request	Result
1	**Map the game**	Suggested a chapter structure built on the settled decisions (G5, G7): one chapter per exam module, Chapter 1 Rules of the Road with eight levels plus a boss, three level types (rapid-fire, steering, plot), and drafts for Chapters 2 to 5.
2	**Chapter 1 mockups**	Drew eleven screens on a design canvas: the chapter map, levels 1 to 8, a miss replay and the boss.
3	**All vessels, spoken misses**	Added the full lights deck (17 patterns, Rules 23 to 30) and a lights-miss screen that says the pattern aloud. Corrected a mix-up: green over white is trawling, not sailing.
4	**Day shapes and a race course**	Added the shapes deck and the Harbor Run course design on the Long Beach prototype. Flagged two errors in outside screenshots (tow lights; red-white-red versus three reds).
5	**Night racecourse prototype**	Built Night Harbor Run: a single-file game with 15 gates, light arcs from Rule 21, a day leg, a fog leg and spoken answers. Opened issue #45 for the repo port.
6	**Level 3 with traffic**	Built the harbor course into /play on branch head-dev/harbor-traffic-course (#46): gates, moving traffic, a wrong turn gets you hit, golden replays, proof bundle.
7	**Chapter 1, level 1**	Built Definitions at /rules on branch head-dev/ch1-definitions (#47): portrait rapid-fire, 17 cards, hearts, replay, stars, mastery record, proof bundle.
8	**Handoff to Grok**	Wrote a self-contained Markdown handoff covering both branches, open questions and the rules still to check.
9	**Gallery page**	Published a page with every image in build order and how game, UX and instructional designers prototype. Fixed a standalone-file bug in Night Harbor Run.
10	**Engineer pack**	Sent a zip with 15 screenshots and the playable HTML file, then this PDF.

3. Product rules the game must keep

From `docs/product/DECISIONS.md`. Build only on settled decisions.

Rule	What it means for engineering
G1, ADR 0001	Strict TypeScript, Phaser 4.2.1 pinned exactly, Vite. src/sim is pure and deterministic: no Phaser, DOM, Math.random, Date or performance.now. Golden replays required.
G6	Every level shows which rule set is in force (Inland, International or both).
G7	Chapter 1 order: definitions, lights, day shapes, horn signals, pecking order, steady bearing, who gives way, fog and narrow channels, boss.
G8	Rapid-fire levels portrait with one thumb; steering levels landscape. Mouse and keyboard on desktop.
G9	One mastery record per rule and situation, shared with the flashcards.
G10	Boss in exam format; pass at the module mark (90% for Rules of the Road) on two runs. uscg-regs checks every question.
G11, G12	Three hull hearts; a miss plays a replay of about 5 s and the situation comes back. 1 to 3 stars; one unlocks the next level.
G15, G16	Interface never uses red or green (they mean port and starboard). Every sound has a visual partner.
A3, guardrails	Game time never counts as sea time. The game never says a player qualifies.

4. The game map

One chapter per exam module (G5). Each chapter is a port on a chart; its levels are buoys along the channel; the boss is the harbor entrance. Three level types repeat across chapters.

Level type	Phone	Teaches by
Rapid-fire	Portrait, one thumb	Fast perceptual classification with instant feedback
Steering	Landscape, twin levers	Applying a rule under time pressure
Plot	Landscape or tablet	Working chart problems (Chapter 5)

Chapter	Exam subject	Level ideas (draft)
1	Rules of the Road	Definitions, lights, day shapes, horn signals, Rule 18 pecking order, Rule 7 steady bearing, Rules 13 to 17, fog and narrow channels, boss
2	Navigation General	Buoy run, compass correction (TVMDC), tides and currents, weather and electronics
3	Deck General	Docking (the Long Beach prototype), anchoring scope, knots, stability, towing
4	Deck Safety	Man overboard turn, fire classes, VHF calls, distress signals, pollution placards
5	Chart Plotting	Plot problems: lat/long, distance, DR, fixes, set and drift



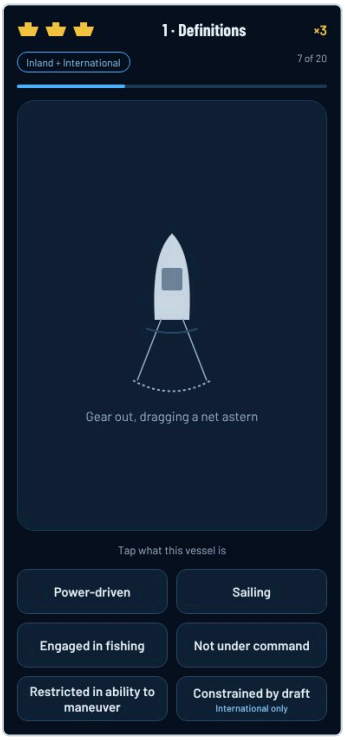
Chapter 1 map: cleared levels gold, next level glowing, boss locked

Proposals still awaiting Et

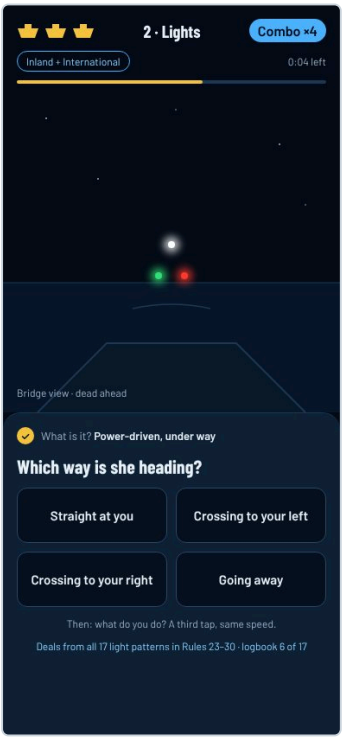
- Clearing Chapter 1 unlocks Chapters 2 to 4 in any order; Chapter 5 after Chapter 2.
- A mixed-review stop every 2 to 3 levels.
- Inland/International "differences" levels after horn signals and after narrow channels.
- Topics that cannot be made fun go to the flashcards instead of a level (G20).

5. Chapter 1 mockups Mockup

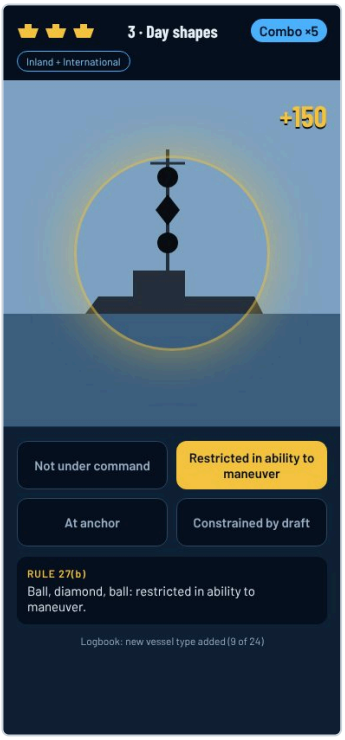
One screen per level. Portrait screens are rapid-fire; landscape screens are steering levels with a lever under each thumb.



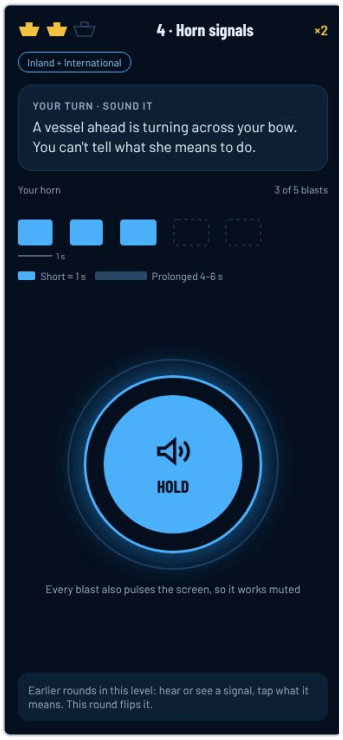
1 · Definitions: tap what the vessel is



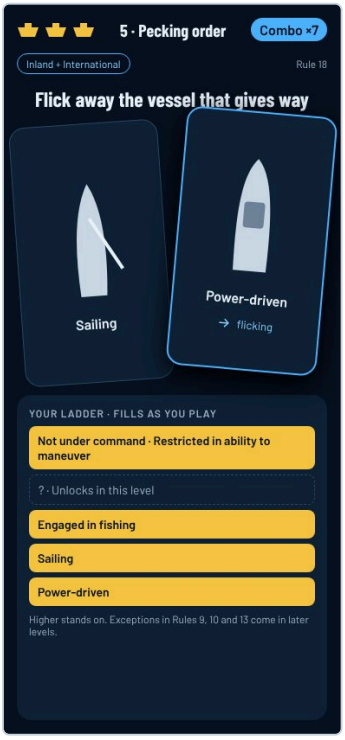
2 · Lights: what is she, which way is she heading



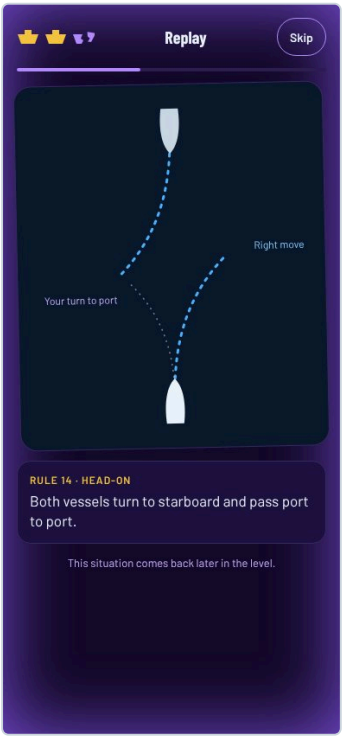
3 · Day shapes: gold flash and rule line on a right answer



4 · Horn: hold to sound, real blast lengths



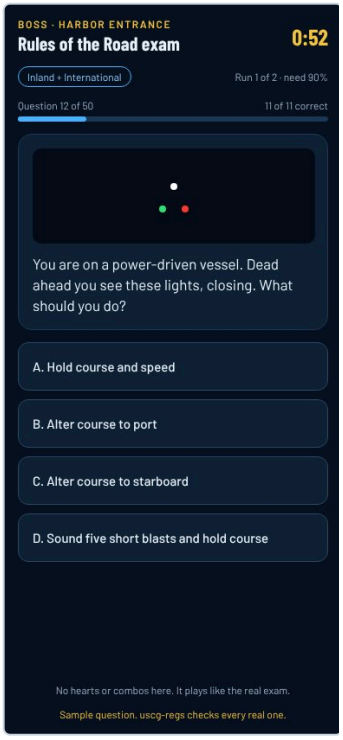
5 · Rule 18: flick away the vessel that gives way



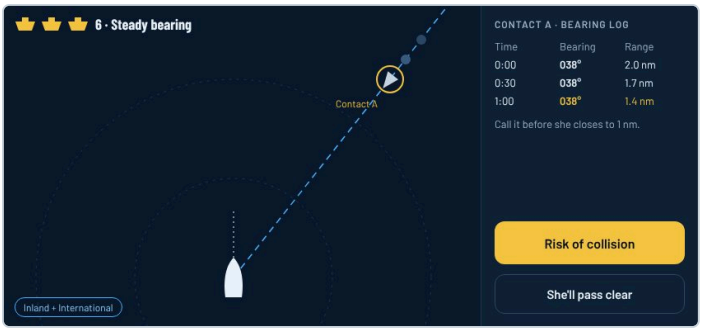
A steering miss: head-on replay, Rule 14



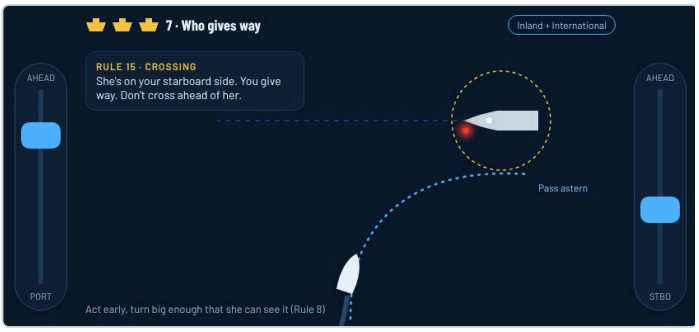
A lights miss: pattern said aloud



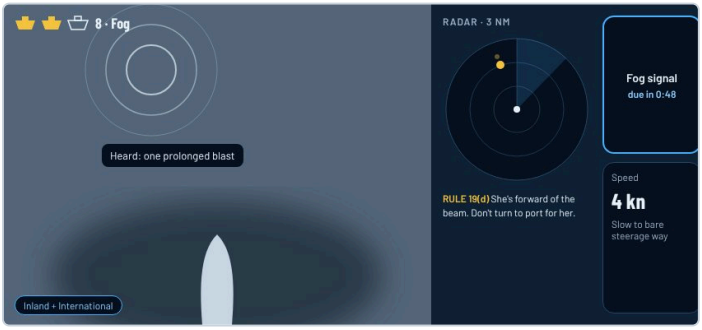
Boss: timed multiple choice at 90%



6 · Steady bearing, Rule 7



7 · Crossing, Rules 13 to 17



8 · Fog, Rule 19

6. Content decks Mockup

Every pattern a level draws from. These lists are what uscg-regs checks.



Lights deck: 17 patterns from Rules 23 to 30. Inland-only patterns go to a later Differences level.

DAY RUN · THE SHAPES DECK

Every ball, cone, diamond and cylinder

By day, these shapes replace the lights. Each one is a boat parked or moving on the Harbor Run course, and the player has to read it at speed and do the right thing.

Draft list. uscg-regs checks every shape and citation against the rule text before it's built.



At anchor
One ball, forward
Rule 30(a)



Not under command
Two balls. She can't get out of your way
Rule 27(a)



Aground
Three balls. Shoal water nearby
Rule 30(d)



Restricted in ability to maneuver
Ball, diamond, ball
Rule 27(b)



Dredging or underwater work
Balls on the blocked side, diamonds on the side you pass
Rule 27(d) · "Pass on the diamonds"



Mine clearance
Three balls in a triangle. Stay 1,000 m clear
Rule 27(f)



Towing, tow over 200 m
A diamond on the tug and on the tow. Never cut between them
Rule 24(a)



Engaged in fishing
Two cones, points together. The hourglass
Rule 26



Fishing, gear out over 150 m
Add a cone point up, on the side the gear runs
Rule 26(c)



Motor-sailing
Cone point down. Sails up, engine on: treat her as power-driven
Rule 25(e)



Constrained by draft
One cylinder. Don't impede her
Rule 28 · International only



Small vessel, divers down
A rigid copy of flag A in place of the shapes
Rule 27(e)

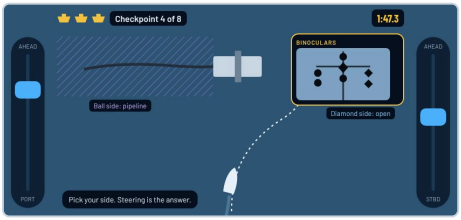
Day shapes deck: every ball, cone, diamond and cylinder, with rule numbers

7. The Harbor Run course design Mockup

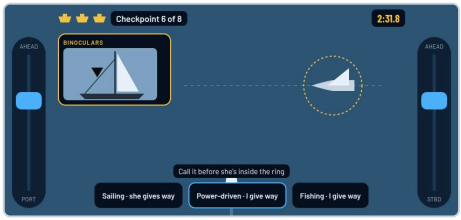
A race course on the Long Beach prototype: eight checkpoints with vessels showing shapes, a dredge fork where steering is the answer, and a docking finish. It became the basis for level 3 (section 9).



Harbor Run: the full course



Pass the dredge on the diamond side



Call the vessel while steering



Finish: splits, stars, race your ghost

8. Night Harbor Run prototype Prototype

One self-contained HTML file (canvas, no backend, desktop and phone). One night lap through 15 gates; each gate triggers an encounter. Rule set: International (COLREGS) throughout. Distances, Rule 22 ranges and fog-signal intervals are compressed; vertical light spacing is exaggerated for phones.

Light rendering (Rule 21)

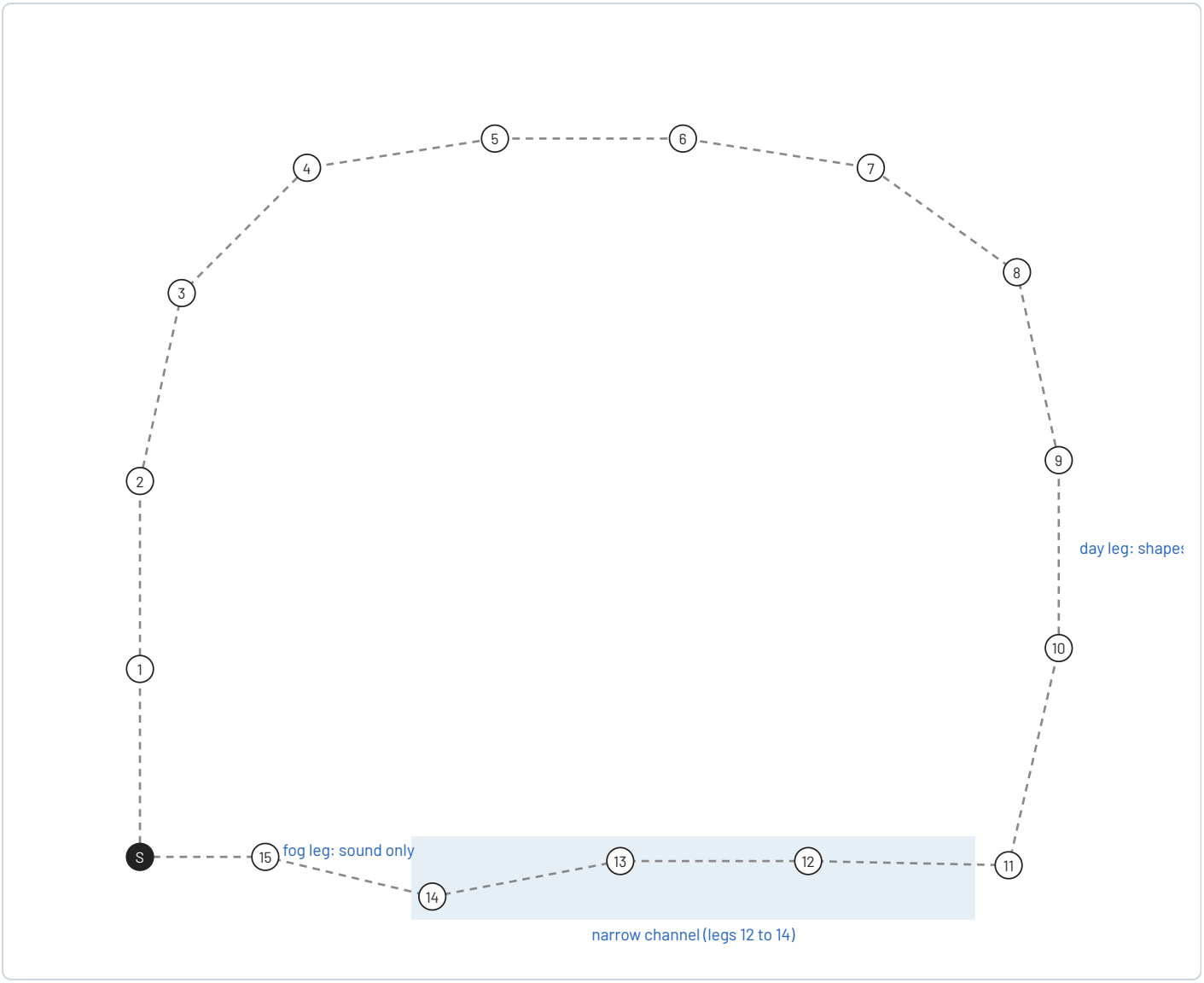
For each light the game computes the observer's aspect: the bearing from the vessel to the player relative to the vessel's heading. A light draws only if its arc covers that aspect: masthead within 112.5 degrees either side of ahead (225 total); starboard sidelight from ahead to 112.5 degrees on the starboard side, port the mirror; sternlight and towing light within 67.5 degrees of astern (135 total); all-round always. Each light is projected from its fore-aft and athwartships offset, so range lights and sidelight spacing show heading, and lights change as the player moves around a vessel.

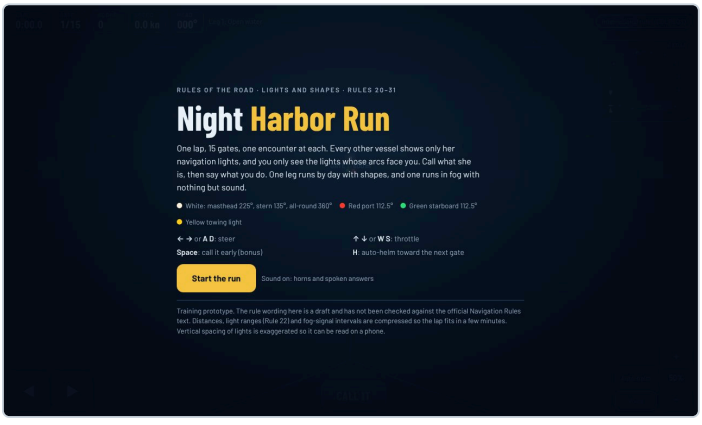
Encounter at each gate

The game pauses (auto at about 270 m, or early with CALL IT for a bonus) and asks: 1 what is she, 2 what do you do, 3 the sound signal where one applies. Feedback shows the right answer, the rule and a short phrase that is also spoken. Collisions, crossing a towline, the dredge pipeline and the shoal cost points.

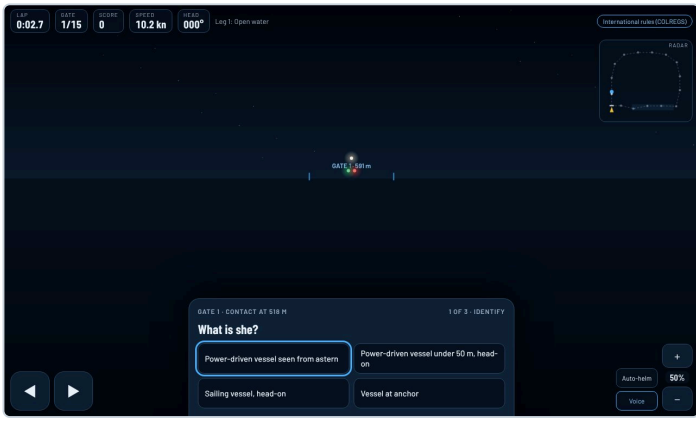
Gate	Encounter	Rules
1	Power-driven under 50 m, head-on	Rules 14, 34(a)
2	Power-driven 50 m or more, crossing from starboard	Rules 15, 16, 34(a)
3	Power-driven crossing from port, five short blasts	Rules 15, 17, 34(d)
4	Overtaking: sternlight only	Rule 13
5	Sailing vessel, red over green	Rules 25, 18
6	Vessel under oars, torch	Rule 25(d)
7	Trawler; gillnetter with a gear light	Rules 26(b), 26(c)
8	Pilot vessel; seaplane	Rules 29, 31, 18(e)
9	Not under command; towboat pushing ahead	Rules 27(a), 24(c), 14
10	Day leg: RAM ball-diamond-ball, anchored ball	Rules 20, 27(b)
11	Aground at night on a shoal	Rule 30(d)
12	Dredge in the narrow channel	Rules 9, 27(d)
13	Constrained by draft, inbound	Rules 28, 9, 18(d)
14	Tug and tow from astern; overtaking signal	Rules 24, 13, 34(c)
15	Fog: one prolonged, two short, unseen	Rules 35, 19

Course layout

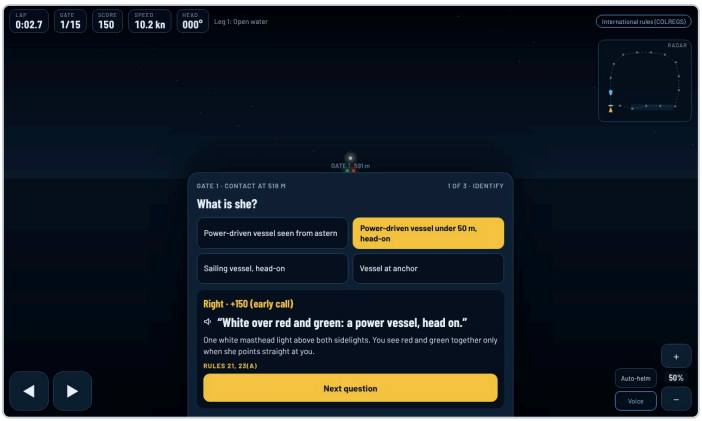




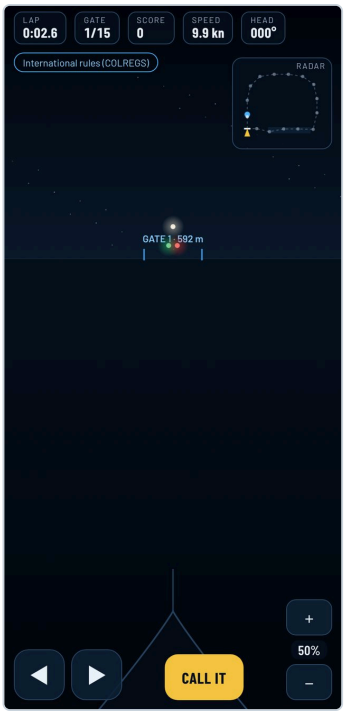
Start screen: light arcs, controls, draft note



Called early: what is she?



Answer with rule and spoken phrase



Phone, portrait

9. Built: level 3 harbor course Built, not merged

Branch `head-dev/harbor-traffic-course`, issue #46. A third level in `/play`: five gates in order (three round anchored boats), then dock. Traffic runs fixed tracks; any contact ends the run with a banner naming the vessel and its lesson. The ferry meets you head-on and turns to its own starboard, so a turn to port steers into it.

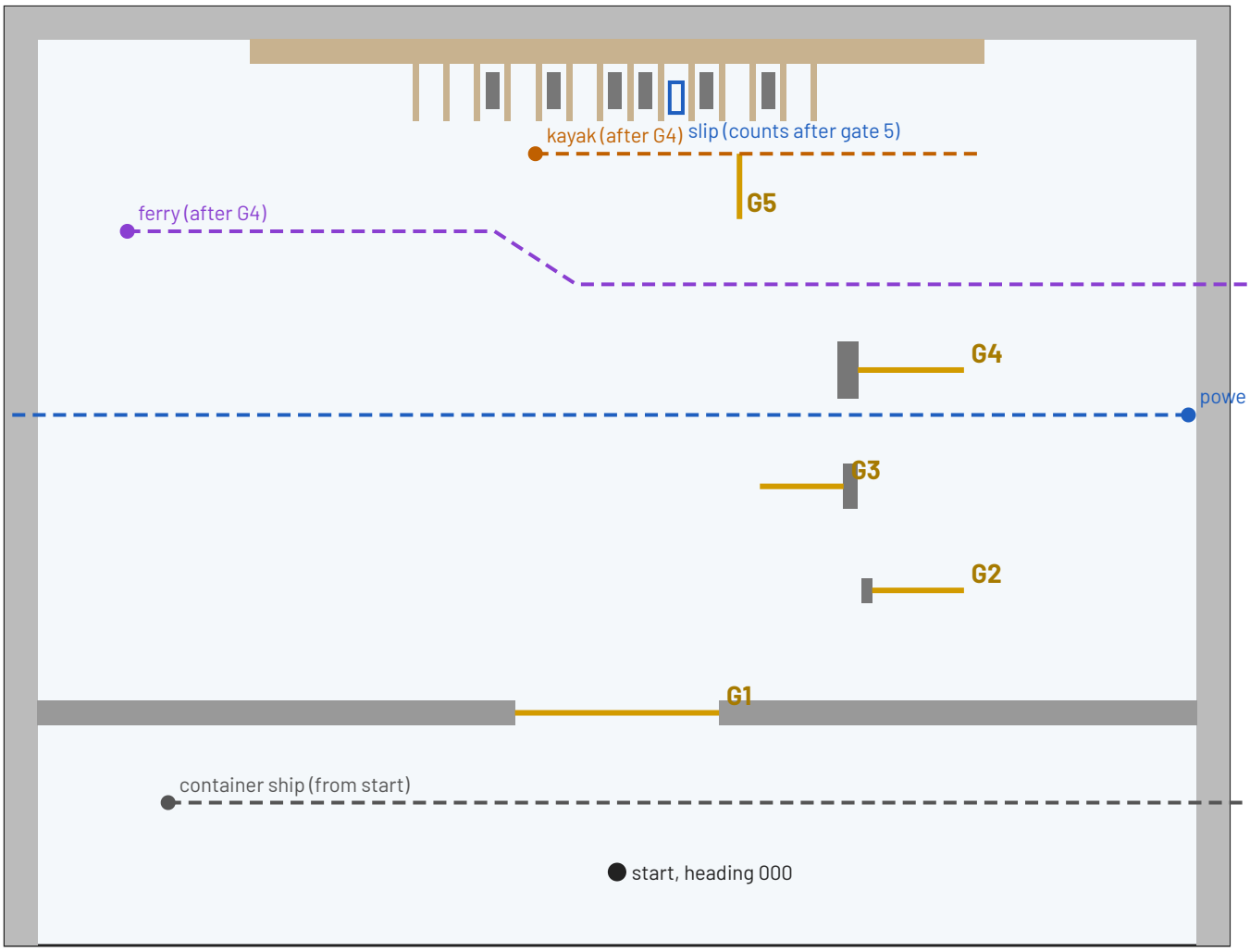
Architecture

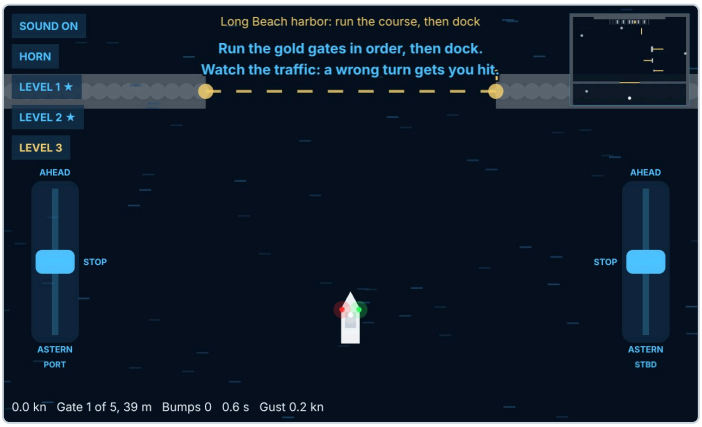
- `src/sim/mission.ts`: optional `gates` (line segments) and `traffic` (name, kind, length, beam, speed, `startAfterGate`, lesson, path) with zod checks.
- `src/sim/course.ts`: integer track poses, hull circles and an integer segment-crossing test. Lengths use `Math.sqrt`, which is correctly rounded in every engine.
- `src/sim/sim.ts`: optional `course` state (gates passed, start frame per vessel, struck-by index). Levels without a course serialize and hash exactly as before; the three older golden replays are byte for byte unchanged. The slip counts only after the last gate.
- `src/game/HarborScene.ts`: gates (gold next, dim passed), top-down traffic with nav lights, chart dots, pointer to the next gate, HUD "Gate n of 5", hit banner with lesson. Debug state adds `xMm`, `yMm`, `gate`, `gates`, `struckBy`.

Vessel	Size (m)	Speed (m/s)	Starts	Lesson on hit (draft)
container ship	60 x 11	3.5	start	A ship in the channel can't stop or turn for you. Stay out of her way (Rule 9).
powerboat	10 x 3.5	4.5	gate 2	She crossed from your starboard side, so you give way: slow down or pass astern (Rule 15).
ferry	30 x 8	4	gate 4	Head-on: both boats turn to starboard and pass port to port (Rule 14).
kayak	4 x 0.9	1.2	gate 4	Small craft are hard to see. Keep a lookout and slow down near them (Rules 5 and 6).

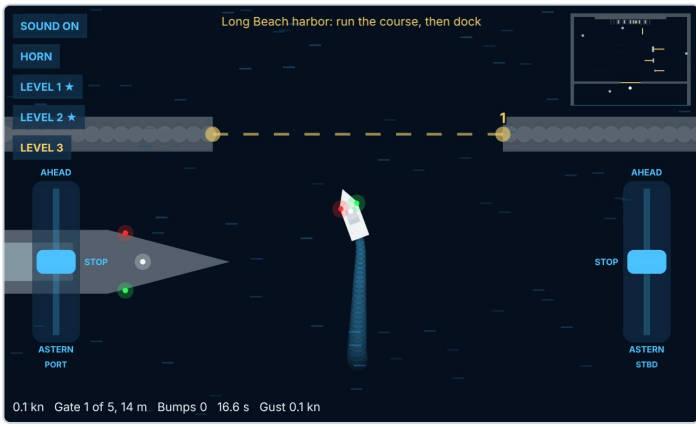
World 300 x 230 m. Par 150 s for three stars. Golden replays: `long-beach-traffic-docks` (a scripted helmsman docks in about 205 s) and `long-beach-traffic-hit-by-ferry` (turns to port, struck by the ferry).

Course layout from the mission file

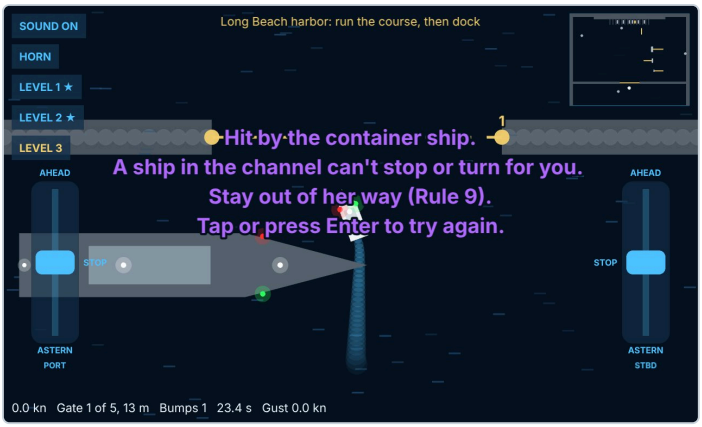




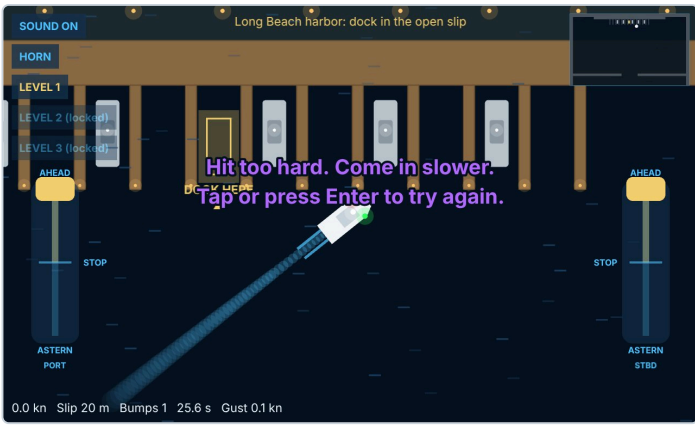
Level 3 start: gate 1 is the breakwater gap



Stopped in the container ship lane



Hit by the ship: name and Rule 9 lesson



Level 1 docking (main), shown for comparison

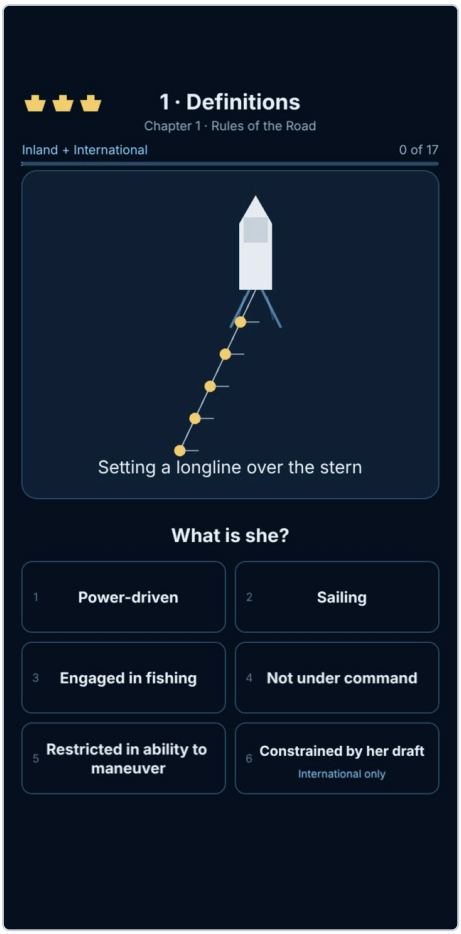
10. Built: Chapter 1, level 1 Definitions Built, not merged

Branch `head-dev/ch1-definitions`, issue #47. A new page at `/rules/`, linked from the front page, offline like the others. Portrait, one thumb; keys 1 to 6 on desktop.

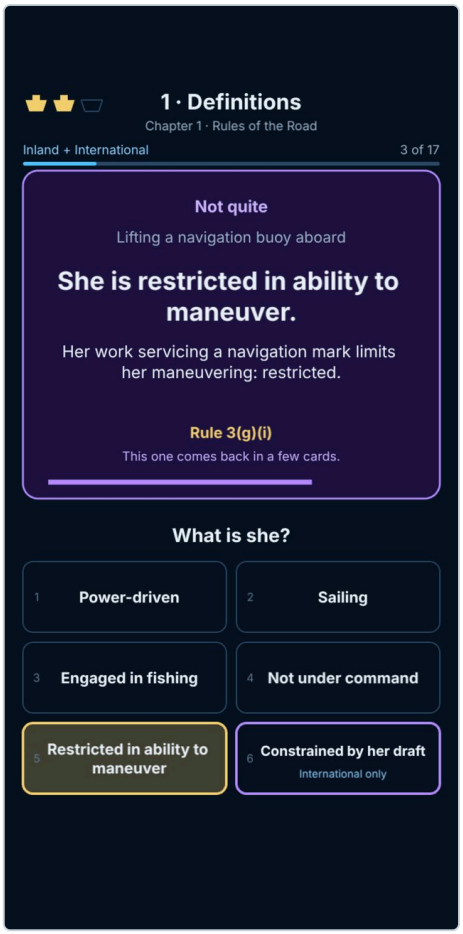
Architecture

- `src/sim/rapid.ts`: pure run logic. Seeded Mulberry32 shuffle; 3 hearts; right answer +100 x combo (max x5); a miss puts the card back 3 cards later; outcome cleared or out-of-hearts; stars: 3 for no misses inside par (75 s, question time only), 2 for one miss, 1 for clearing.
- `src/sim/chapter1/definitions.ts`: 6 categories and 17 cards (id, art, cue, answer, rule, reason, rule set).
- `src/game/rapid/DefinitionsScene.ts` and `vesselArt.ts`: Phaser scene and top-down drawings; `src/game/storage.ts` adds a mastery record per card id.

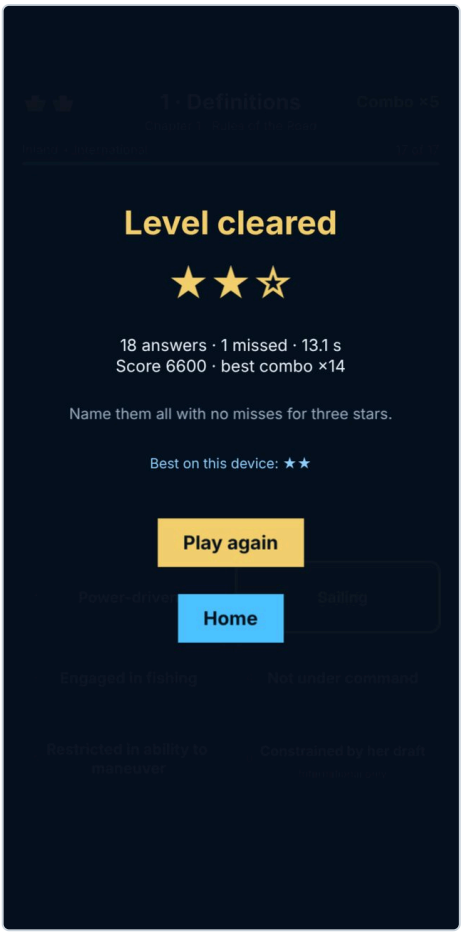
Card id	Cue	Answer	Rule
`rule3:motorboat`	Motoring across the harbor	Power-driven	Rule 3(b)
`rule3:container-ship`	Container ship under way at sea	Power-driven	Rule 3(b)
`rule3:sailing`	Sails full, engine off	Sailing	Rule 3(c)
`rule3:spinnaker`	Racing downwind under spinnaker	Sailing	Rule 3(c)
`rule3:motorsailing`	Sails up, engine running	Power-driven	Rule 3(c)
`rule3:sails-furled`	Sails furled, motoring home	Power-driven	Rule 3(b)
`rule3:trawler`	Towing a trawl net astern	Engaged in fishing	Rule 3(d)
`rule3:longliner`	Setting a longline over the stern	Engaged in fishing	Rule 3(d)
`rule3:trolling`	Trolling two lines for salmon	Power-driven	Rule 3(d)
`rule3:engine-failed`	Engine failed, drifting	Not under command	Rule 3(f)
`rule3:steering-failed`	Steering gear broken, circling	Not under command	Rule 3(f)
`rule3:buoy-tender`	Lifting a navigation buoy aboard	Restricted in ability to maneuver	Rule 3(g)(i)
`rule3:dredge`	Dredging the channel	Restricted in ability to maneuver	Rule 3(g)(ii)
`rule3:divers`	Tending divers working below	Restricted in ability to maneuver	Rule 3(g)(ii)
`rule3:heavy-tow`	Heavy tow: she can't turn off her course	Restricted in ability to maneuver	Rule 3(g)(vi)
`rule3:replenishment`	Transferring cargo to another ship, under way	Restricted in ability to maneuver	Rule 3(g)(iii)
`rule3:deep-draft`	Deep tanker in a narrow channel, little water under her	Constrained by her draft	Rule 3(h), International



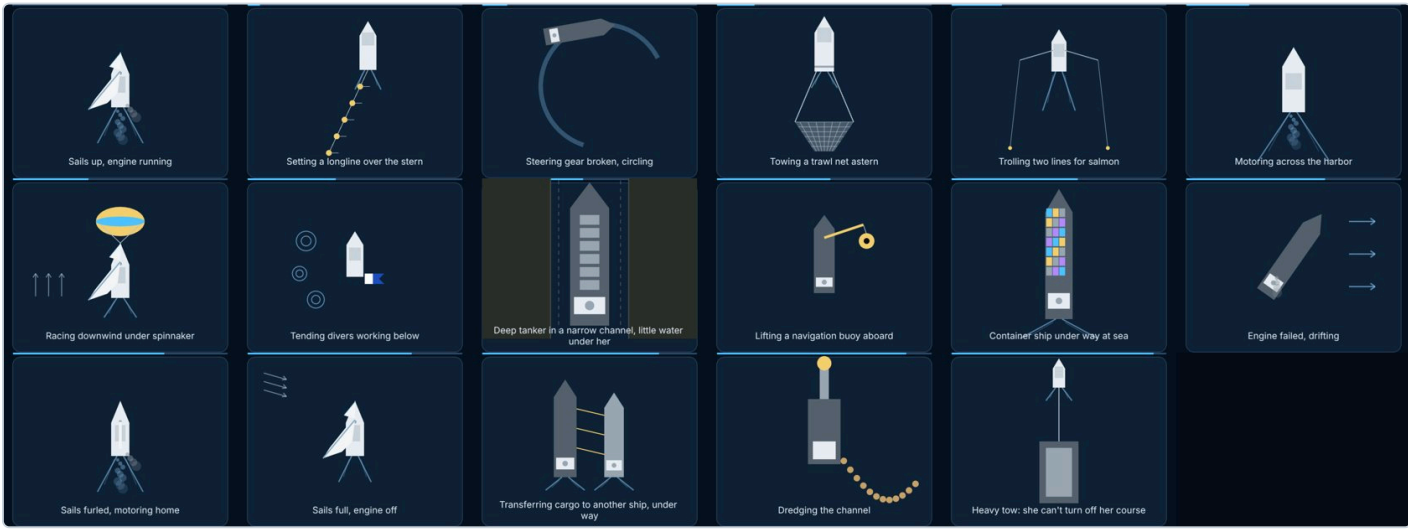
A card, phone upright



A miss: heart lost, 5 s replay with the rule

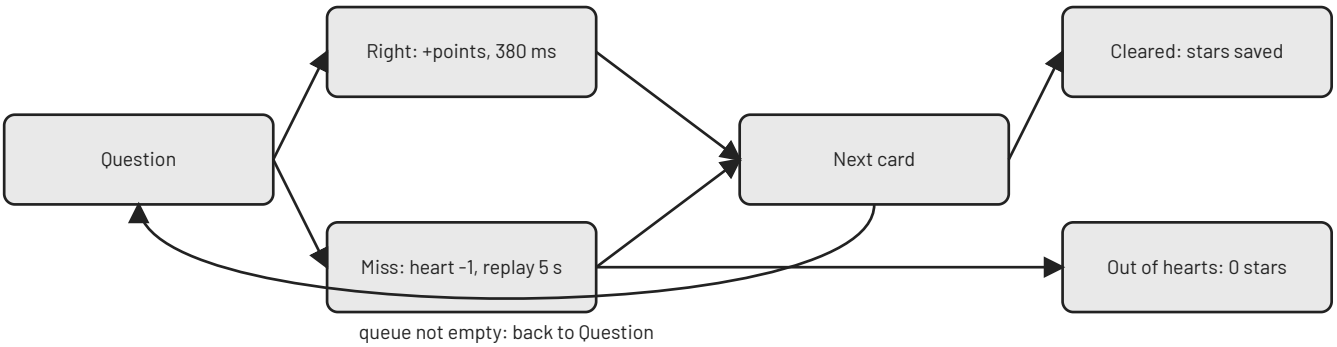


Cleared with two stars



All 17 vessel drawings

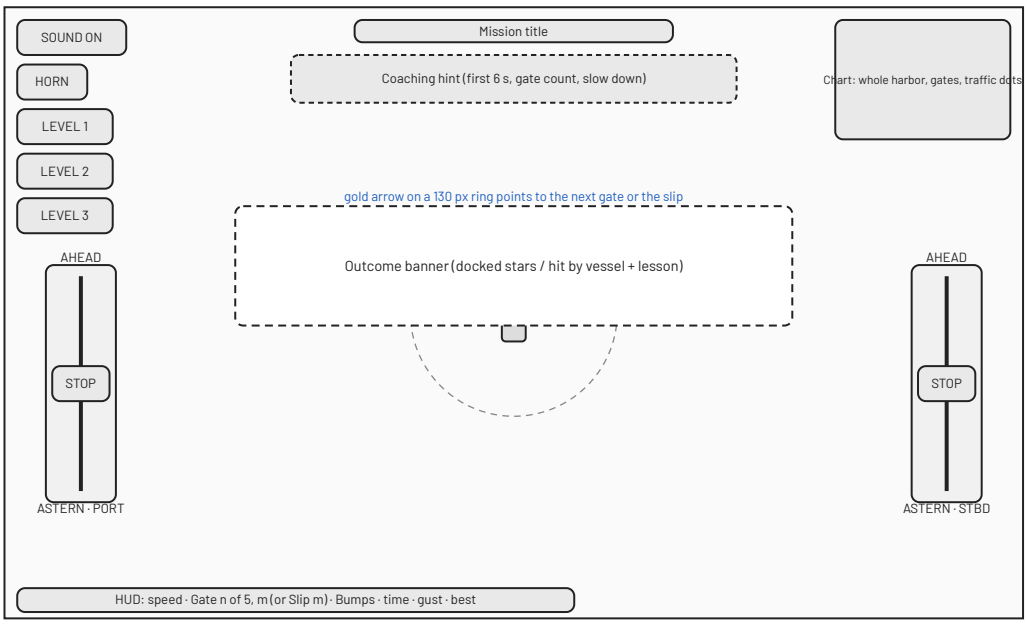
Screen flow


Question
Right: +points, 380 ms
Miss: heart -1, replay 5 s
Next card
Cleared: stars saved
Out of hearts: 0 stars
queue not empty: back to Question

11. Wireframes with layout numbers

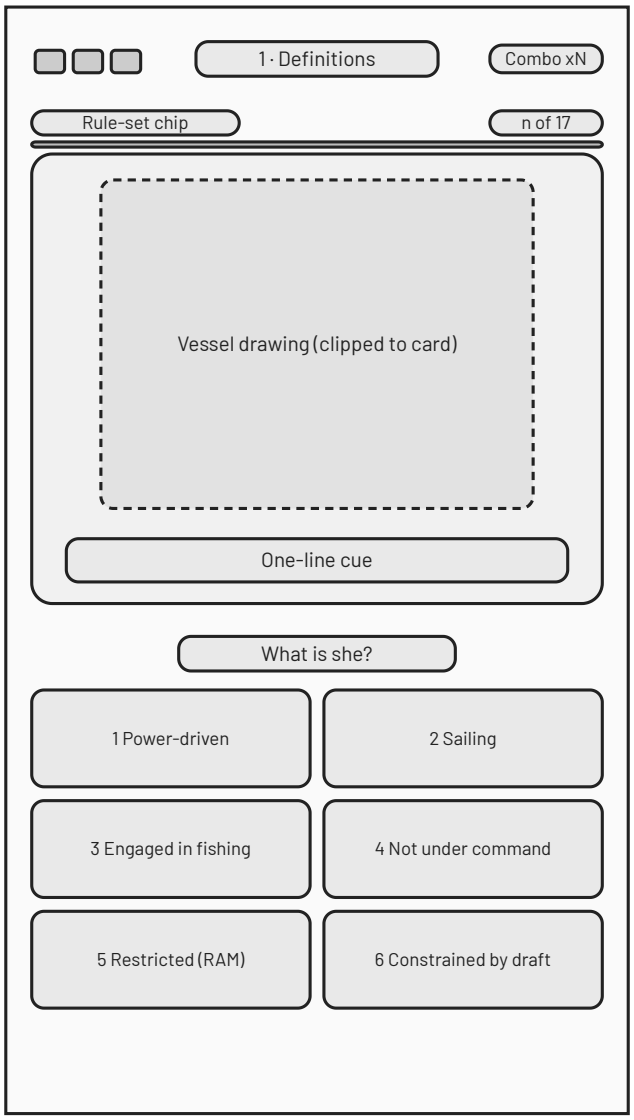
Grayscale wireframes drawn from the coordinates in the code. Blue notes are the numbers an engineer needs.

/play docking and course levels (landscape)



Game space 1280 x 768, Phaser Scale.FIT
Levers: x 96 and x 1184
Stop at y 473, track 338 to 608 (#46)
(on main: 298 to 648)
Level buttons x 16, y 128 + 56 n
Keys: W/S or arrows both levers,
A/D split, Q/Z port, E/C stbd, X stop,
H horn, M sound, Enter retry
Portrait phone: rotate prompt, sim paused

/rules rapid-fire level (portrait)



Game space 720 x 1280, portrait, Scale.FIT

Card: x 30, y 170, 660 x 520

Buttons: 322 x 112, gap 16,

columns x 30 and x 368, first row y 790

Button n centre: (191 | 529, 846 + 128 row)

Keys 1 to 6 answer; Enter skips replay

Miss: purple replay over the card, 5 s,

tap skips after 0.8 s; right button gold,

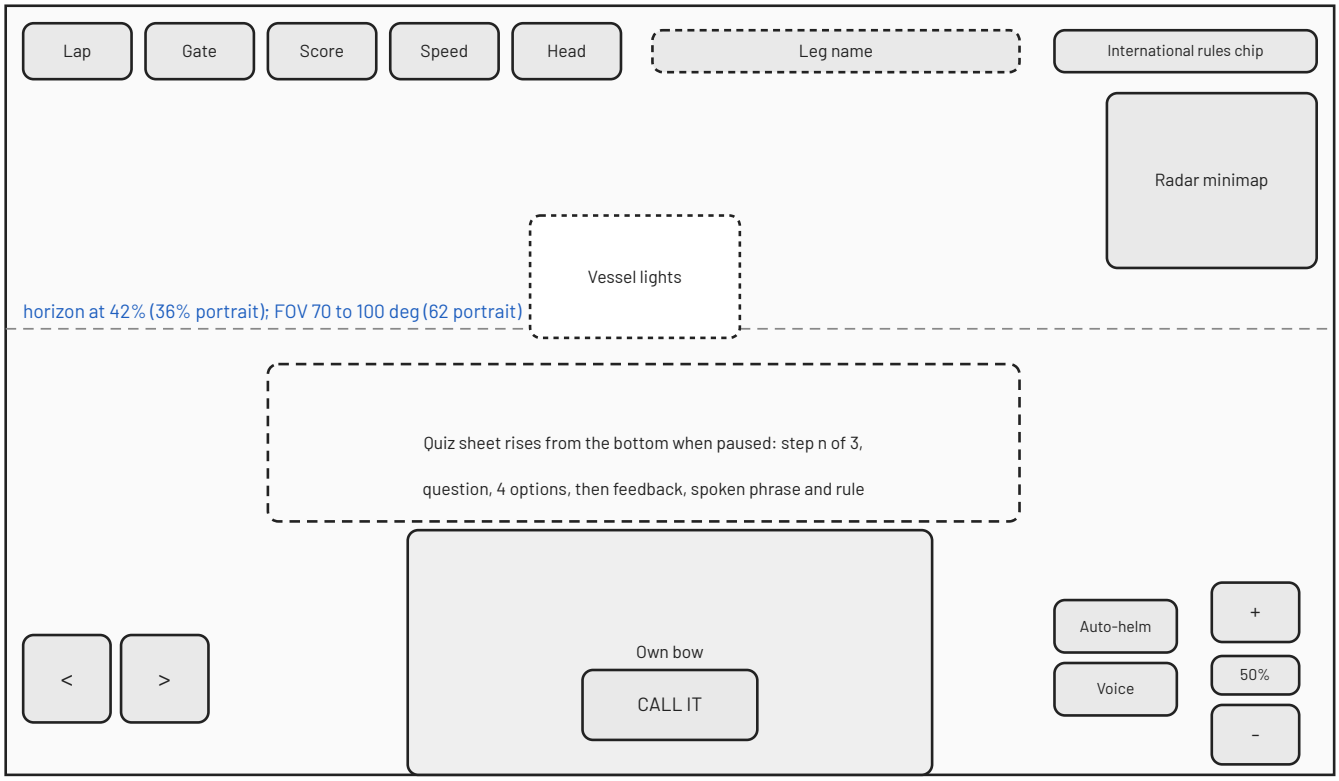
wrong pick purple; card returns 3 later

Right: +100 x combo (max x5), 380 ms pause

End: overlay with stars, Play again, Home

State: window.__captainsAcademyQuiz

Night Harbor Run (landscape; stacks on a phone)



12. Verification and how to run it

1. Install with `npm ci`. If the lockfile must change, use `npx npm@11.21.0 install` (npm 10 fails on this tree).
2. Run `bash scripts/verify.sh`: npm ci, tsc with strict flags, ESLint with import boundaries, Prettier, mission validation, Vitest with golden replays, vite build, Playwright on desktop, phone landscape and phone portrait, npm audit. It passed on both branches.
3. Drive the real build with `.claude/skills/verify-captains-academy/control.sh launch | doctor | drive STEPS.json | stop`. Recipes: `harbor-course.steps.json` (#46) and `definitions.steps.json` (#47). The driver now opens other pages (`"path"`), waits on `gt` / `lt` , and taps right or wrong answers at /rules.
4. Every PR commits a PASS proof bundle from `scripts/write-proof.sh`; CI runs `scripts/check-proof.sh`. Agents clock in and out on time cards in `agent-resources/timecards/`.
5. Golden replay files must not change unless meant to: `VERIFY_UPDATE_REPLAYS=1 bash scripts/verify.sh`.

Test	Covers
tests/course.test.ts	Gate crossing, traffic tracks and start triggers, slip waits for last gate, traffic hit at rest, ferry replay
tests/replays.test.ts	Five golden replays, older three unchanged
tests/rapid.test.ts	Shuffle determinism, combo, miss return, hearts, stars, deck integrity, answer key in step with the deck
e2e/play.spec.ts	Level 3 opens after level 2 and counts gate 1
e2e/rules.spec.ts	Clean run with 3 stars, a miss and its replay, out of hearts; all three device projects
e2e/home.spec.ts	Front page has three links

13. Open questions, rules to check, corrections

Waiting on Et (issue #45)

1. Sequencing: A3 and G19 put the sea-time log before the full game. Building Chapter 1 now was taken as Et's call; confirm.
2. Rule set: International only, or an Inland variant too?
3. Audience: license candidates, Coast Guard Academy cadets, or both?
4. Boss: does the race course replace the G10 multiple-choice boss, or run alongside it?
5. Whether to open pull requests for #46 and #47.

Rules content still to check (uscg-regs, against 33 CFR 83 and COLREGS)

- All 17 Definitions cards: answers, reasons, Rule 3 citations.
- Level 3 traffic lessons citing Rules 5, 6, 9, 14 and 15.
- Night Harbor Run answers, especially the rowboat (Rule 18 does not list vessels under oars) and the seaplane (Rule 18(e)).
- G10a: NMC confirmation of exam numbers and a Coast Guard reviewer before the OUPV boss and launch.

> **Corrections made during the session.** Green over white is trawling; a sailing vessel shows sidelights and sternlight and may add red over green. Three white masthead lights are on the towing vessel (tow over 200 m), not the vessel being towed. Red-white-red is restricted in ability to maneuver; three reds is constrained by draft, International only.

Known limits

- Night Harbor Run and the mockups are not in the repo; the HTML file is in the zip.
- Level 3 golden replay helmsman is slow (about 205 s against a 150 s par), so a skilled player beats it.
- The lever tracks on #46 are shorter (half-travel 135 instead of 175) to fit LEVEL 3.

14. How experts prototype, and the next steps

Game designers

1. Paper prototype
2. Greybox: real controls, no art
3. Vertical slice: one level at final quality
4. Weekly playtests with strangers
5. Production, then polish

UX designers

1. Research the hardest real user
2. Sketches and wireframes
3. Mockups
4. Clickable prototype
5. Usability tests with a handful of users per round
6. Developer handoff

Instructional designers

1. Analyze exam outcomes
2. Design objectives and measures
3. Develop in short cycles (SAM)
4. Subject-expert review
5. Pilot and evaluate pass rates

Stage	Status
Analyze, plan, mockups, prototype	Done
Vertical slice (level 3, Definitions)	Done, not merged
Playtest with real learners	Not done
Expert rules review	Not done

Recommended next steps

1. Playtest Definitions and level 3 with about five license candidates on their own phones; watch, do not help.
2. Run the uscg-regs check on every card, lesson and citation.
3. Fix what both find; test again.
4. Then build Chapter 1, level 2 (Lights): portrait rapid-fire on `src/sim/rapid.ts`, a bridge view drawing only lights whose Rule 21 arcs face the player, the 17-pattern deck, and a spoken phrase on every miss.

15. Appendix: files, branches and links

Item	Path or link
Repository	https://github.com/Etdev2/Captains-academy-
Issues	#45 Night Harbor Run · #46 level 3 harbor course · #47 Definitions
Branch #46	`head-dev/harbor-traffic-course`: `src/sim/course.ts`, `src/sim/sim.ts`, `src/sim/mission.ts`, `missions/long-beach-traffic.json`, `src/game/HarborScene.ts`, `src/game/helm.ts`, `tests/course.test.ts`, `tests/replays/long-beach-traffic-*`
Branch #47	`head-dev/ch1-definitions`: `src/sim/rapid.ts`, `src/sim/chapter1/definitions.ts`, `src/game/rapid/*`, `rules/index.html`, `tests/rapid.test.ts`, `e2e/rules.spec.ts`
Verification skill	`.claude/skills/verify-captains-academy/` (features `harbor-course.md`, `definitions.md`)
Engineer zip	captains-academy-screens.zip: 10 arcade screenshots, 5 Night Harbor Run screenshots, night-harbor-run.html
Claude artifacts (private to Et)	Chapter 1 mockups canvas · Night Harbor Run · gallery page "Rules of the Road Build Log"
Handoff for Grok	handoff-grok.md, sent in the session

Generated 22 October 2026 from the repository and the session's own files. Screenshots come from the production build driven by the verification skill, or from the rendered mockups.